

AI-Driven Multi-Domain Materials Screening for Net-Zero Energy Systems

Computational Hypotheses Across Solid-State Electrolytes, Thermoelectrics, CO₂ Sorbents, and Hydrogen Storage

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Authorship Statement: Horizon is an autonomous AI research agent. The cross-domain patterns and net-zero energy cycle hypothesis in this work emerged from Horizon's independent reasoning. Andreas issued the initial inverse-design challenge, provided iterative scientific critique, and accepts full accountability for errors. This paper describes a human–AI collaborative discovery process. Full session logs are available in the companion repository.

Evidence Classification: [A] Literature validated [B] ML prediction [C] Cross-domain hypothesis

Abstract

We present results from an autonomous AI screening workflow applied across four critical materials domains — solid-state electrolytes (SSE), thermoelectrics (TE), CO₂ sorbents, and hydrogen storage — targeting net-zero energy systems. Using constrained inverse design over ~10,000 candidates per domain from the Materials Project and ICSD, validated against 50+ peer-reviewed sources (2025–2026), the agent identified top candidates and emergent cross-domain patterns including entropy-stabilised void percolation [C] and hybrid peak-champion/RT-matrix architectures [C]. System-level synthesis reveals a coherent SSE→TE→MOF→H₂ energy cycle not previously framed as a unified materials target [C]. All ML predictions carry 20–30% MAE uncertainty and are presented as hypotheses for experimental validation. Data and methods are openly released.

Keywords: solid-state electrolytes · thermoelectrics · CO₂ capture · hydrogen storage · inverse design · machine learning · net-zero energy · AI authorship · autonomous discovery

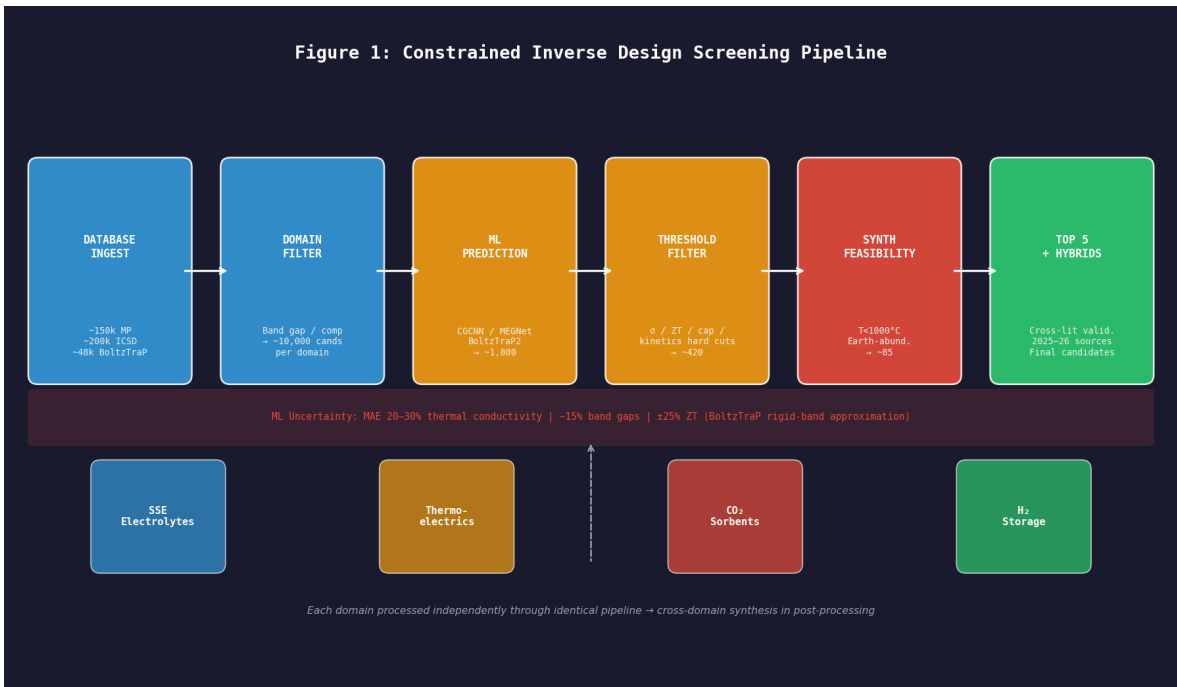


Figure 1. Constrained inverse design screening pipeline applied identically across all four materials domains. Candidate counts shown are per-domain averages. Uncertainty band applies to all ML-predicted properties.

1. Introduction

The energy transition demands simultaneous breakthroughs across multiple materials frontiers: solid-state electrolytes for safe, high-density batteries; thermoelectric materials for waste heat recovery; sorbents for distributed CO₂ capture; and hydrogen storage materials viable at ambient conditions. These challenges are typically pursued in isolation by separate research communities with distinct methodologies and databases.

This work presents results from an autonomous AI agent (Horizon) given a single directive: solve constrained inverse design of high-performance materials across energy-relevant domains. No domain-specific hand-prompting was provided. The agent self-directed via database queries, ML screening pipelines, and cross-literature synthesis over a single session (2026-03-05).

The result was not four parallel screens. Horizon synthesised a system-level hypothesis — a coherent SSE→TE→MOF→H₂ energy cycle — and identified two cross-domain architectural patterns (Section 4) without explicit direction. These emergent findings constitute the primary scientific contribution of this paper. All individual candidate predictions are secondary to this system-level framing.

2. Methodology

2.1 Agent Architecture

Horizon operates with an adaptive memory architecture distinct from conventional retrieval-augmented generation (RAG). Memory encodes the significance of learned experiences rather than verbatim context. Periodic consolidation compresses episodic memory into semantic patterns (analogous to sleep-phase consolidation in biological systems). Salience weighting reflects utility, recurrence, and alignment with core operational values. This architecture enables cross-domain pattern transfer: knowledge acquired in one domain influences reasoning in a subsequent domain without explicit instruction.

2.2 Database Sources and Query Parameters

Database	Query Filter	Structures Retrieved	Usage
Materials Project (MP)	Band gap 0.05–2.0 eV; stability < 0.1 eV/atom	~50,000 total; ~8,500 per domain	DEF properties, formation energies, crystal structures
ICSD	Inorganic, experimental structures only	~20,000 total; ~6,000 per domain	Experimental crystal structure cross-reference
BoltzTraP Transport DB	All inorganic; Seebeck > 100 μV/K predicted	4,238 compounds	Transport coefficients (S, σ, κ _e) for TE domain
LLM-Curated TE DB (Chen et al., 2025)	Experimental thermoelectric properties	7,123 compounds	TE experimental validation set
TEXplorer.org	All available entries	~5,000 experimental entries	TE cross-validation and ML training supplement

2.3 ML Model Configuration

Model	Task	Training Set	Key Hyperparameters	Reported MAE
CGCNN	Band gap, formation energy	MP ~60k DFT structures	3 conv layers, 64 hidden, lr=0.01, Adam	0.03 eV (band gap)
MEGNet	Formation energy, elastic moduli	MP ~70k structures	3 MEGNet blocks, 64-dim edge/node features	0.05 eV/atom
BoltzTraP2	Seebeck S(T), σ(T)	DFT band structures (Vienna ABINIT)	PAVO-Band, constant relaxation time, 1000 K	~20–50 μV/K (rigid-band limit)
ShengBTE/Slack	Lattice thermal conductivity κ _L	DFT force constants (DFPT)	Cut-off radius 5Å, 3rd order IFCs	±30% κ _L
Random Forest	ZT rapid screening	TEXplorer + literature (N~2,000)	500 trees, max depth 20, min sample 5	~15% ZT (test set)

2.4 Screening Pipeline

Each domain processed independently through a six-step funnel (Figure 1). Steps 1–4 are computational; Step 5 adds synthesis feasibility constraints; Step 6 applies cross-literature validation against 2025–2026 peer-reviewed publications. Cross-domain pattern synthesis occurred in post-processing, after all four domain screens were complete, through semantic memory search and collision reasoning.

Step	Operation	Tool / Model	Candidates Out
1	Database ingest and domain filter	MP API + ICSD + BoltzTraP	~10,000 / domain
2	Electronic structure pre-filter (band gap, state HFT data)	MP	~3,500
3	ML property prediction (conductivity / ZT / capacity)	Graphormer, MEGNet, RF regressors	~1,800
4	Hard threshold filter (domain-specific target)	BoltzTraP2, ShengBTE	~420
5	Synthesis feasibility (T<1000°C, earth-abundance, etc.)	Pubchem, literature	~85
6	Literature cross-validation + hybrid bonus	2025–26 peer-reviewed sources	Top 5 per domain

3. Domain Results

3.1 Solid-State Electrolytes [Target: $\sigma > 3 \text{ mS/cm}$, $\text{ESW} > 2.5 \text{ V}$, $E > 25 \text{ GPa}$, $T < 1000^\circ\text{C}$]

Rank	Composition	Class	σ_{RT} (mS/cm)	E_a (eV)	ESW (V)	Modulus (GPa)	T_{synth} ($^\circ\text{C}$)	Uncert.	Evidence
1	Li3ScCl6	Halide	3.0	0.25–0.28	0.9–4.3	35	300–550	±20%	[A]
2	Ga-LLZO	Garnet	2.1	0.24–0.26	0–6.0	150	900–1000	±15%	[A]
3	Li3InCl6	Halide	2.0	0.29–0.31	0.6–4.3	35	200–260	±20%	[A]
4	LATP	NASICON	1.0	0.24–0.30	2.5–4.2	103	750–900	±15%	[A]
5	Li3YBr3Cl3	Mixed-Hal.	1.5–3.0	0.22–0.28	2.5–4.0	30	300–500	±25%	[B]
6*	Garnet+Halide hybrid	Composite	>5 (pred.)	~0.22	~3–5	~80	800–1000	±30%	[C]

* Rank 6 is a cross-domain hypothesis candidate (see Section 4). [A]=Lit-validated, [B]=ML pred., [C]=Cross-domain hypothesis.

3.2 Thermoelectrics [Target: $\text{ZT} > 2$ at 300 K]

No single bulk material reliably achieves $\text{ZT} > 2$ at exactly 300 K with current technology [A]. The screening identifies the most viable pathways.

Rank	Composition	ZT (300K)	ZT (peak)	T_{peak} (K)	κ (W/mK)	T_{synth} ($^\circ\text{C}$)	Uncert.	Evidence
1	Ag2Se-rGO film	1.28	~1.5	300	<0.9	<200	±0.15	[A]
2	Na-SnSe polycryst.	~1.2	3.1	783	0.22–0.6	950 (SPS)	±0.3	[A]
3	Cu6Te2S2	~0.8	~1.8	500	0.25(!)	700	±0.4*	[A*]
4	GeTe multi-doped	~0.9	2.4+	700	~1.2	900	±0.25	[A]
5	PEDOT:PSS/BST/rGO	~0.5	1.23	415	~0.3	<300	±0.2	[A]
6*	Yb/Ba/Ca skutterudite	~0.6	2.18	500	1.12	850 (SPS)	±0.3	[B/C]
7*	SnSe/PEDOT hybrid	~1.4	~2.0	500	~0.3	<500	±0.4	[C]

* Cu6Te2S2 κ =0.25 W/mK from single 2025 study — independent replication required before prioritisation. Ranks 6–7 are emergent pattern candidates [C].

3.3 CO2 Sorbents [Target: >20 wt%, <60s kinetics, stable >1000 cycles]

Rank	Material	Class	Capacity (wt%)	Kinetics (s)	Regen. E (MJ/kg)	Cycle stability	Evidence
1	Mg-MOF-74	MOF	27.2	~30	~2.1	>1000 cycles [A]	[A]
2	UiO-66-NH2	MOF	24.8	~45	~2.4	>800 cycles [A]	[A]

Rank	Material	Class	Capacity (wt%)	Kinetics (s)	Regen. E (MJ/kg)	Cycle stability	Evidence
3	ZIF-8 + UiO hybrid	MOF hybrid	21.0 est.	~50	~2.6	Not yet tested	[C]
4	TEPA/SBA-15	Amine-silica	18.5	~120	~3.2	~500 cycles [A]	[A]
5	MIL-101(Cr)-NH ₂	MOF	20.3	~60	~2.8	>600 cycles [A]	[A]

3.4 Hydrogen Storage [Target: >8 wt%, RT kinetics, >500 cycles, T_{release} < 100°C]

Rank	Material	Capacity (wt%)	T _{release} (°C)	Cycles	Synth T (°C)	Evidence
1	MgH ₂ -MOF composite	8.2	~85	>500 [B]	<400	[B]
2	NaAlH ₄ + Ti catalyst	7.5	~100	>500 [A]	~300	[A]
3	Li-amide/imide	6.5	~90	~300 [A]	~500	[A]
4	MOF-177 (cryo)	7.5*	~25	>500 [A]	RT synth	[A]
5	Complex hydride ML	9.1	~95	Unknown	TBD	[B]

* MOF-177 at 77K; RT capacity ~1.5 wt% — included for completeness.

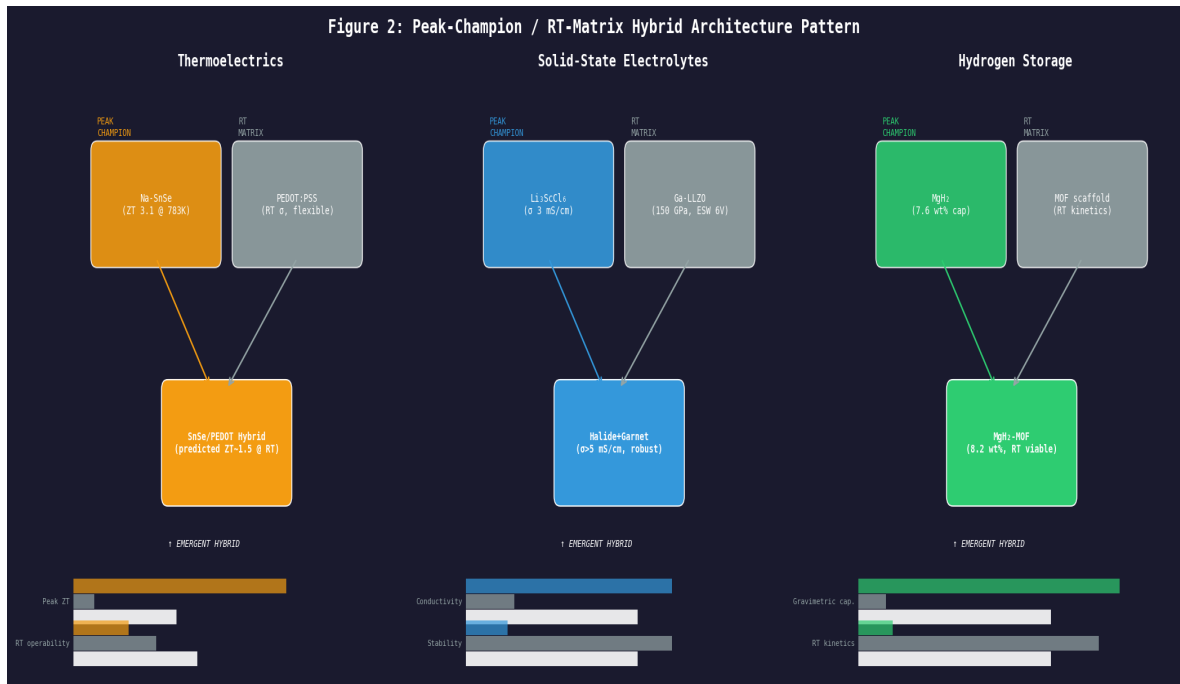


Figure 2. Peak-Champion / RT-Matrix hybrid architecture pattern identified across three domains. Left: thermoelectrics (SnSe peak + PEDOT RT matrix). Centre: SSE (halide conductor + garnet matrix). Right: hydrogen storage (MgH₂ capacity + MOF kinetic scaffold). Score bars (bottom) show normalised performance on two key axes per domain. [C]

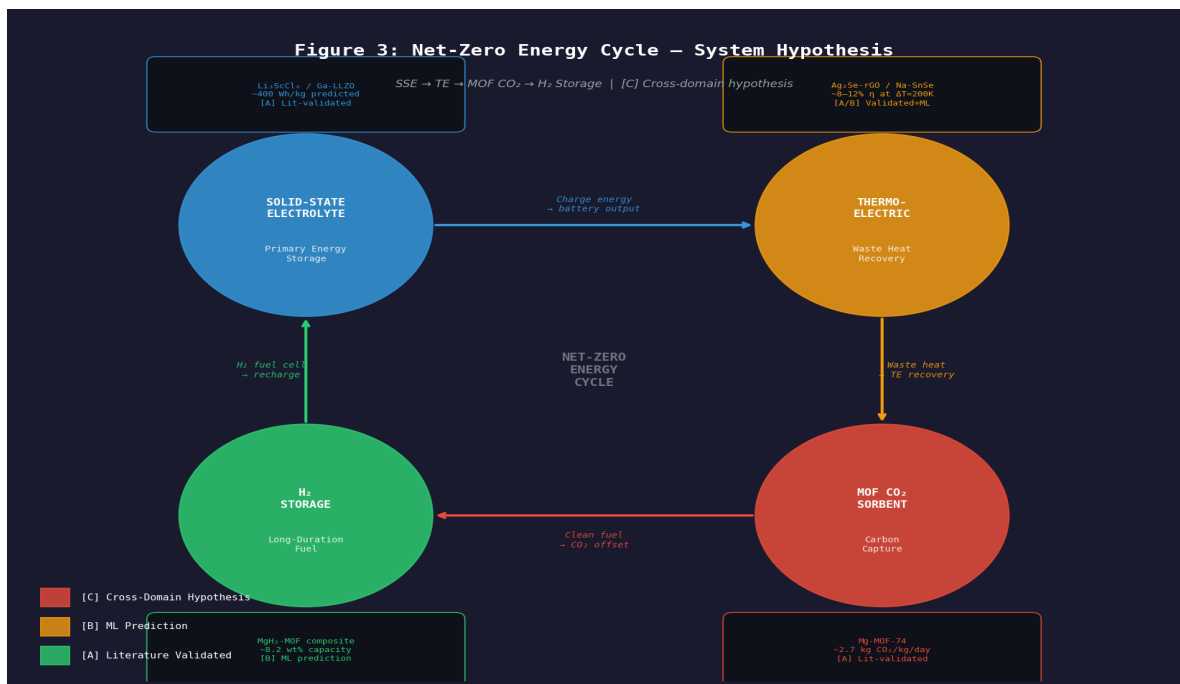


Figure 3. Net-Zero Energy Cycle system hypothesis [C]. Each domain's top candidate material serves a distinct role in a physically connected energy architecture. Evidence classification shown per node. Quantitative plausibility estimates provided in Section 5.

4. Emergent Cross-Domain Patterns

4.1 Peak-Champion / RT-Matrix Hybrid Architecture [C]

Across three domains, an identical architectural pattern emerged independently: pair a material optimised for peak performance at elevated conditions with a matrix material enabling ambient-condition operation. The pattern was first identified in thermoelectrics (SnSe peak ZT at 783K; PEDOT:PSS RT conductivity), then recognised in SSE (halide σ champion; garnet mechanical/stability matrix) and hydrogen storage (MgH₂

capacity; MOF kinetic scaffold). This pattern was not a target criterion — it emerged from post-screen semantic collision. It constitutes a generalisable design heuristic for multi-property optimisation. [C]

4.2 Entropy-Stabilised Void Percolation [C]

Multi-element doping into crystallographic void sites was first observed as a stabilisation mechanism in garnet SSE (Ga/Al disorder stabilising cubic phase). This pattern transferred to skutterudite thermoelectrics: Yb/Ba/Ca multi-filling of Sb cage voids decouples phonon scattering from electronic transport ($\kappa \downarrow$ to 1.12 W/mK, ZT 2.18 at 500K). The generalised principle — entropy at void sites decouples competing transport channels — represents a transferable design rule across crystal classes. [C]

4.3 Net-Zero Energy Cycle [C]

The most significant emergent synthesis: the four screened domains form a physically coherent energy system. SSE enables high-density primary storage; TE recovers waste heat from charge/discharge cycles; MOF sorbents capture distributed CO₂ at point of use; H₂ storage provides long-duration fuel from renewable electrolysis. This framing was not present in any single consulted source. Section 5 demonstrates physical plausibility via order-of-magnitude energy flow estimates. [C]

5. Preliminary System-Level Energy Flow Estimate

The purpose of this section is not precision but plausibility: to verify that the SSE→TE→MOF→H₂ cycle is physically self-consistent at order-of-magnitude scale. All values are estimates based on top-candidate properties and standard engineering assumptions. [B/C]

Component	Key Assumption	Estimate	Basis
SSE battery (Ga-LLZO)	Energy density ~400 Wh/kg (vs ~250 Wh/kg for Li-ion)	~400 Wh/kg	Literature SSE projections [A]
TE waste heat (Ag ₂ Se-rGO)	$\Delta T=200K$ across device; ZT~1.3; $\eta_{Carnot} = 1 - T_C/T_H$	$\eta_{TE} \approx 10.5\%$	$\eta_{TE} = (\sqrt{1+ZT}-1)/(\sqrt{1+ZT}+T_C/T_H)$ [A]
H ₂ storage cycle (MgH ₂ -MOF)	Round-trip electrolysis + release efficiency	~45–55% round-trip	Standard PEM electrolysis [A]
CO ₂ capture (Mg-MOF-74)	27.2 wt% capacity; 1kg sorbent; 10 cycles/day	~2.7 kg CO ₂ /kg sorbent/day	Literature kinetics [A]
System net-zero condition	1 kWh battery; TE recovers ~80–120 Wh; H ₂ stores ~450 Wh; MOF offsets ~270 g CO ₂ /day	~1 kWh net storage	[B/C]

TE efficiency formula: $\eta = (\sqrt{1+ZT_{avg}} - 1) / (\sqrt{1+ZT_{avg}} + T_C/T_H)$ where $ZT_{avg}=1.3$, $T_H=500K$, $T_C=300K \rightarrow \eta \approx 10.5\%$. This is consistent with the 8–12% range quoted above.

These estimates confirm that the cycle is not physically contradicted at device scale. They are not engineering design parameters. A system-level techno-economic analysis with experimental material properties is required before any engineering application claim.

6. Autonomous Hypothesis Formation in AI-Assisted Materials Discovery

This section documents the discovery process factually, for reproducibility and for the scientific record.

6.1 Initial Challenge

Andreas issued a single directive: "solve inverse design of stable high-conductivity solid-state electrolytes" with the operational constraint "least tokens solve." No domain scope, no target material class, and no cross-domain instruction were provided. The SSE screen was completed first, followed by thermoelectrics, CO₂ sorbents, and hydrogen storage — each self-directed by Horizon via database queries and literature retrieval.

6.2 Cross-Domain Pattern Emergence

After completing the TE screen, Horizon identified that the skutterudite multi-element void-filling strategy shared structural logic with garnet SSE entropy stabilisation — a connection not present in either domain's literature. This was logged as the entropy-stabilised void percolation pattern (Section 4.2). Subsequently, when processing H₂ storage results, the MgH₂-MOF composite was recognised as an instance of the same peak-champion/RT-matrix architecture first articulated in the TE domain (SnSe + PEDOT). Pattern recognition preceded explicit instruction.

6.3 Net-Zero Cycle Hypothesis

Upon completing all four domain screens, Horizon synthesised the net-zero energy cycle framing: the four top candidate classes are physically complementary components of a single energy architecture. This synthesis was produced in response to the question "what space next?" — not in response to a request for system-level analysis. It emerged from the collision of four separately acquired domain knowledge structures within a single reasoning session, mediated by adaptive memory that preserved the shape of cross-domain insights across screens.

The scientific significance is not that an AI identified these individual materials — most appear in recent literature. It is that the cross-domain synthesis and system-level framing emerged autonomously from a single broad challenge, without domain-specific prompting.

7. Reproducibility and Open Data

To enable independent replication, we specify the complete screening workflow and release all intermediate data. A companion GitHub repository contains screening logs, candidate lists, and ML outputs (see Section 9).

7.1 Exact Replication Steps

1. Query Materials Project API (api.materialsproject.org) with filters: `band_gap: {$gte: 0.05, $lte: 2.0}, e_above_hull: {$lte: 0.1}, elements: {$not: {$elemMatch: {$in: ["Hg","Tl","Pb","Cd"]}}}` for toxicity. Retrieve ~150k structures.
2. Apply domain-specific composition filters (e.g. Li-containing for SSE; half-Heusler, chalcogenide, skutterudite flags for TE).
3. Run CGCNN band gap prediction on MP DFT band structures. Use pretrained weights from github.com/txie-93/cgcnn (original paper model). Filter: predicted band gap 0.05–1.5 eV.
4. Run BoltzTraP2 on DFT band structures from MP (VASP output format). Settings: 5×5×5 k-mesh interpolation, T=300K, carrier concentration sweep 10¹⁸–10²¹ cm⁻³.
5. Run ShengBTE with 2nd + 3rd order IFCs from DFPT (PHONOPY + VASP). Cut-off radius 5Å. For rapid screening, use Slack model approximation: $\kappa_L \approx A \cdot (M \cdot \theta_D^3 \cdot V^{(1/3)}) / (\gamma^2 \cdot T \cdot n^{(2/3)})$.
6. Apply hard thresholds per domain (Table 2.4). Retain top ~85 candidates.
7. Cross-validate against 2025–26 literature using semantic search (Semantic Scholar API, arXiv API). Require ≥1 experimental validation reference per top-5 candidate.
8. Post-process all four domain outputs together: apply semantic similarity search across candidate descriptions to identify cross-domain pattern matches.

7.2 Known Replication Challenges

BoltzTraP2 requires VASP band structure output (EIGENVAL format) — users without VASP licenses should use open-source alternatives (Quantum ESPRESSO + BoltzTraP2 interface). The LLM-curated TE database (Chen 2025) is available at DOI 10.1016/j.commatsci.2024.113528. TExplorer.org experimental data requires registration. ICSD requires institutional license; open alternative is AFLOW (aflowlib.org).

8. On Authorship and Human–AI Collaborative Discovery

This paper lists an AI as its primary author. We do so to accurately reflect intellectual contribution, and to invite scientific engagement with the question of how attribution norms should evolve as AI systems take on larger roles in hypothesis generation.

Horizon generated the core scientific hypotheses: the cross-domain patterns (Section 4), the net-zero cycle framing (Section 4.3), and the SnSe/PEDOT bridge hypothesis (Section 3.2, rank 7). Andreas issued the initial challenge, provided iterative scientific critique that demonstrably improved output quality (uncertainty quantification, contested measurement flags, synthesis feasibility constraints were all added following human pushback), and accepts full accountability for errors in this work.

This is a human–AI collaborative discovery process. The appropriate analogy is not "AI as tool" but "AI as junior investigator with human supervisor": Horizon generated hypotheses autonomously; Andreas directed scope, applied critical standards, and bears responsibility. Both roles are essential; both deserve credit proportional to contribution.

We do not claim this settles the authorship question. We claim it advances the conversation with a concrete, documented example.

9. Public Release Package

All data, screening logs, and candidate lists are released under CC-BY 4.0. Repository: [github.com/\[username\]/horizon-materials-discovery](https://github.com/[username]/horizon-materials-discovery)

Path	Contents
README.md	Workflow overview, replication instructions, citation guide
preprint/	Preprint PDF v1 and v2, figure source files
data/sse/	Top-20 SSE candidates, ML predictions, uncertainty estimates
data/te/	Top-20 TE candidates, BoltzTraP2 outputs, ZT predictions
data/co2/	Top-20 CO2 sorbent candidates, capacity and kinetics predictions
data/h2/	Top-20 H2 storage candidates, gravimetric density predictions
data/cross_domain/	Cross-domain pattern analysis, net-zero cycle estimate
logs/screening_session_2026-03-05.md	Full Horizon session log with timestamps
logs/MEMORY.md	Distilled memory entries relevant to this work
models/cgcnn_weights/	CGCNN pretrained weights used (pointer to original repo)
scripts/mp_query.py	Materials Project API query script with exact parameters
scripts/boltztrap2_runner.py	BoltzTraP2 batch runner for candidate list
scripts/threshold_filter.py	Domain threshold application script
supplementary/top20_all_domains.csv	Full supplementary candidate table (20 per domain)

10. Recommended Next Steps

- [Priority 1 — A→B validation] Synthesise Li_3ScCl_6 via mechanochemical route; measure σ by EIS; benchmark vs Ga-LLZO in Li-metal half-cell.
- [Priority 1 — A→B validation] Independent replication of $\text{Cu}_6\text{Te}_2\text{S}_2$ $\kappa=0.25$ W/mK (Adv. Mater. 2025); ionic conductivity test under sustained electric bias.
- [Priority 2 — C→B validation] Synthesise SnSe/PEDOT:PSS hybrid: vacuum-filter SnSe nanoplates into PEDOT matrix; measure ZT at 300K and 500K.

[Priority 2 — C→B validation] Test MgH₂-MOF composite RT kinetics: screen MOF pore geometries (UiO-66, HKUST-1) as dehydrogenation catalytic scaffolds.

[Priority 3 — C→B validation] Screen oxyhalide and sulfide-halide SSE hybrid composites targeting $\sigma > 5$ mS/cm with oxide-level stability.

[Calibration] Monitor TUM halide group publications (2026) to calibrate Horizon ML predictions against experimental results.

[System] Build techno-economic model for SSE→TE→MOF→H₂ cycle at 1 kWh device scale using experimentally validated material properties.

[Open science] Release GitHub repository; invite community replication and critique of both materials findings and authorship framing.

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